Remote Sensing Image Acquisition, Analysis and Applications

Module One

Acquiring images and understanding how they can be analysed

Solutions to end-of-lecture quizzes

John Richards
The University of New South Wales
The Australian National University

Lecture 1. What is remote sensing?

- What are the benefits of imaging from space?
  A large field of view is possible, allowing large regions of the earth's surface to be seen in a single image. It is also feasible to image the whole of the earth's surface in a practical time frame. We will see in a later lecture that spacecraft are above the atmosphere and thus are not affected by atmospheric turbulence.

- What are the benefits of imaging from aircraft?
  Higher spatial resolutions are generally possible, and the user can usually control the region to be imaged. Often, the imaging characteristics (e.g. the wavebands in which imaging takes place) can be changed between missions allowing more flexibility than with satellite programs.

- What are the disadvantages of using an aircraft to image the earth's surface?
  Generally, it is more expensive per region imaged to use an aircraft than to use of satellite data which is shared over 100s to 1000s of end users, allowing the imaging costs to be amortised over a much larger user base. Aircraft also fly through the atmosphere, which is turbulent, meaning that there can be distortions in the geometry of the recorded images.

- Why should we want to form images of the earth's surface?
  Images are similar to maps but can reveal, when analysed, an enormous amount of information about the natural cover types on the surface of the earth, how they change with time, how humans have altered and used the landscape, how we might localize searches for minerals and other resources, and as means for creating and up-dating maps.

- The atmosphere can selectively (with wavelength) absorb radiation from the sun, and from the earth to the sensor. Would you expect the atmosphere to affect imaging using aircraft more or less than imaging from satellites?
  Since aircraft fly closer to the earth's surface the radiation reaching their sensors will travel though a much smaller atmospheric column than the radiation travelling to the sensors on satellites. One would expect, therefore, that the atmosphere would have less effect on the images recorded by aircraft.

Lecture 2. The atmosphere

- Mobile (cell) phones operate at wavelengths of about 1GHz to 2GHz. Would you expect mobile telephone reception to be affected by atmospheric absorption?
  Looking at the graph of the transmission of the atmosphere we see that the atmosphere is almost, but not quite, totally transparent at those frequencies. We

would expect very little effect of the atmosphere on mobile phone reception. That is particularly the case because the distances involved between a mobile handset and a nearby cell base station is generally only in the order of 10's of kilometres, or less.

- Is atmospheric absorption a problem for us when we image the ground from an aircraft?
  This is essentially a repeat of a question from the last lecture, just to remind us that aircraft remote sensing missions are not significantly affected by atmospheric absorption. Flying through the atmosphere is another matter—aircraft encounter turbulence, which leads to geometry distortions in recorded imagery. By contrast, this is less of a problem for satellites.

- Sensors for monitoring fires on the earth's surface tend to operate at about 3-5mm. Is the atmosphere a problem for effective imaging?
  Again, looking at the graph of atmospheric transmission we see there is some significant absorption at those wavelengths. When monitoring fires from aircraft, which is currently the most common situation, the atmospheric column is rarely large enough to cause serious problems.

Lecture 3. What platforms are used for imaging the earth's surface?

- Would a satellite with a 12-noon descending node be good for mapping landscape features?
  No, because we depend upon shadowing to help us see topographic detail in the landscape. Shadowing is minimal at noon, but better early to mid-morning.

- Landsat 7 takes 16 days to image the whole earth surface. How many orbits does it make in that time? Its orbital period is 98.9 minutes.
  There are 1440 minutes per day, so there will be almost 15 orbits per day; USGS works on 14. Over 16 days that gives 224 orbits. If the exact figure of 14.56 orbits per day is used then there will be 233 orbits over 16 days.

- What platform would you choose in each of the following applications?

  - Mapping forests on a continental scale.
    Satellite

  - Daily monitoring fence conditions on a farm.
    Drone (UAV)

  - Monitoring an actively burning forest fire that covers about 50ha.
    Aircraft

  - Seasonal monitoring of crop growth and condition in large scale agriculture
    Satellite

Lecture 4. How do we record images of the earth's surface?

- CCD array sensors tend to have higher spatial and spectral resolution than those based on scanning mirrors. Why?
  Scanning sensors allow each "pixel" to be seen for only a short time during the across-track sweep, compared with a push broom CCD array which allows each "pixel" to be seen for a longer time. The reason for this is that the scanning mirror has to cover ***all*** the pixels in a scan line compared with the CCD, which has one detector per pixel across the scan. Because of the longer observation (or dwell) time per pixel, more energy per unit area is received by the CCD sensor allowing finer spatial resolutions to be achieved and narrower spectral bands, if needed. We can look at this mathematically, in an approximate manner, as follows.
  The spectral power density from the sun created at the earth's surface can be regarded as a constant at a given time. It has units of watts per square metre per micrometre of wavelength ($Wm^{-2}\mu m^{-1}$). From this density field the actual energy at the ground captured by a sensor, with a given pixel size and over a particular band can be expressed as
  Energy in joules = spectral power density ($Wm^{-2}\mu m^{-1}$) x pixel size (m^2) x width of the spectral band (μm) x dwell time (sec)
  Thus, if the dwell time is much larger, as in the case of a CCD array, the pixel size and width of the spectral band can be smaller for the same energy capture.

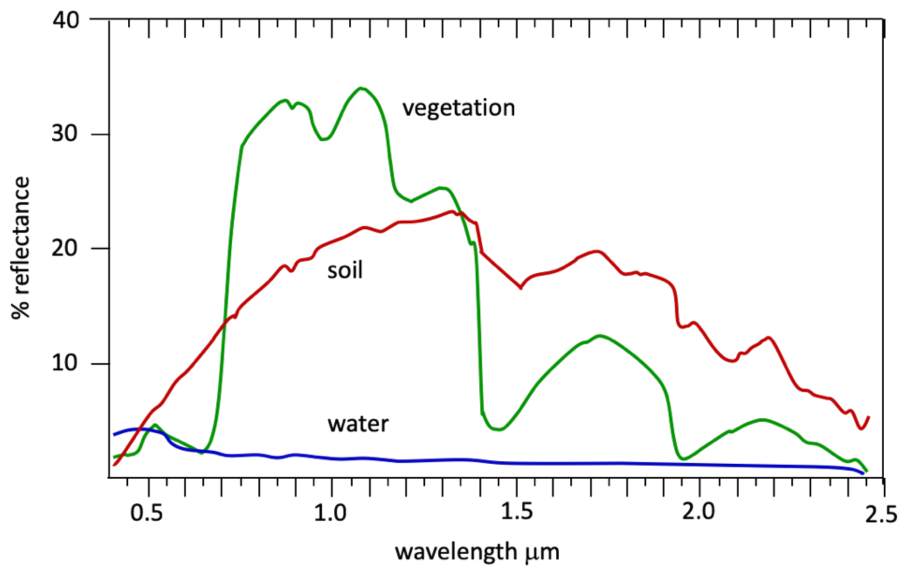
- With aircraft systems, why is IFOV a more meaningful term than pixel size? Has it to do with the flying height?
  Pixel size is defined by the product of IFOV (in radians) and the altitude of the sensor (in metres). Since aircraft fly at different altitudes in different missions the concept of spatial resolution, or pixel size on the ground, is not helpful because it can be different from mission to mission. By contrast IFOV is a constant from mission to mission since it is a characteristic of the sensor and not the platform. Satellites fly at a constant altitude so it is meaningful to talk in terms of spatial resolution, or pixel size.

- Will pixel size vary across the swath for an image recorded by a remote sensing platform?
  Yes, because the IFOV of the sensor is a constant. The pixel size on the ground is defined by the product of the IFOV and the distance from the sensor to the spot on the ground. That distance is smallest directly under the platform (at nadir) and is largest at the swath edges. Thus, the pixel areas imaged across the swath gets larger from nadir to the swath edge. For many sensors (e.g. on Landsat) that can be a negligible difference, but for wide FOV sensors, such as that on Aqua and Terra, the effect is quite large.

Lecture 5. What are we trying to measure?

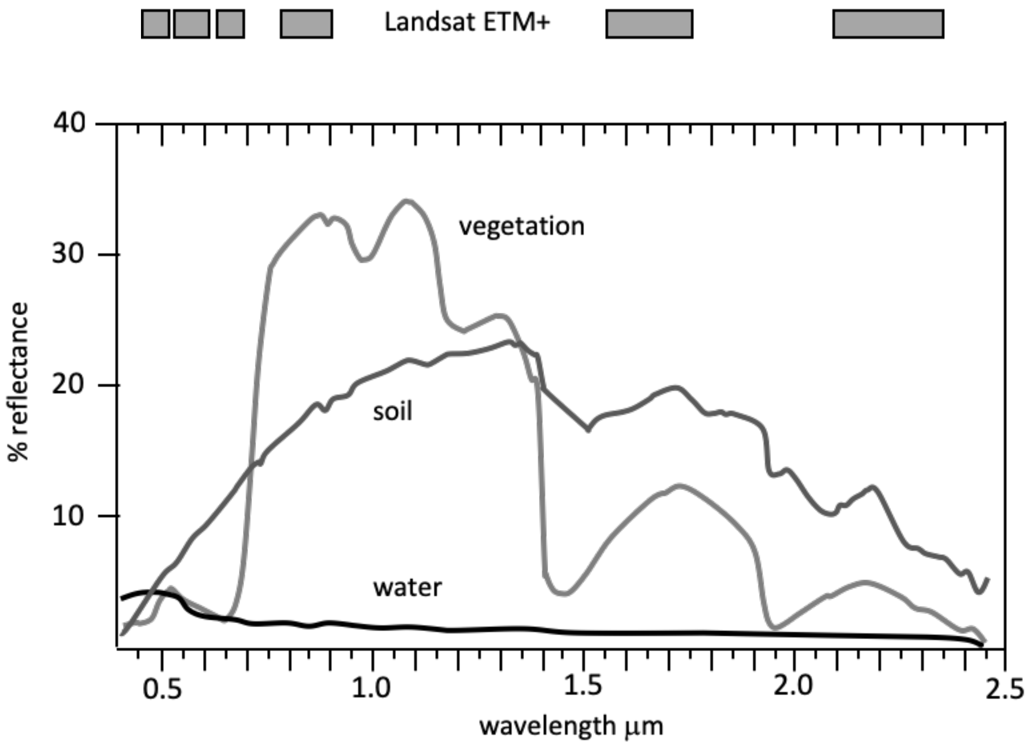
- What wavelength would you choose if you had to image a burning wildfire on the earth's surface? Would it show any useful detail in regions where there is no fire?
  The Planck radiation curve has a maximum in the vicinity of 3-5μm for a black body at a temperature of 1000K, which is about the temperature of a well-developed wildfire. A sensor operating at those wavelengths would be best for detecting burning fires, and would certainly receive a much greater signal than optical or thermal sensors. For non-fire regions the level of detail in a 3-5μm image would not be good.

- What single wavelength would you choose (approximately) if you wanted to produce an image with best discrimination among water, soil and vegetation features?
  To answer this, we return to the spectral reflectance curves of common earth surface cover types.



If the soil were as depicted here, i.e. not bright like sand, a near-IR wavelength would be good. However, often soil and vegetation are both fairly bright in the near-IR. While that would give good differentiation from water, often a visible red band would be chosen for good vegetation/soil discrimination. Water would generally be slightly darker than vegetation.

- Explain why, in your opinion, the Landsat ETM+ bands are placed where they are.
  The following diagram is repeated from the lectures and shows the placement of the ETM+ bands.



The first three bands sample the blue, green and red portions of the spectrum (important to water and vegetation studies), while the fourth samples the near-IR, which we know is especially important in vegetation studies; it is affected by the cellular structure of vegetative matter. The other two sample into the mid-IR but avoid the water absorption bands. That region is important to the detection of mineralogically important soils and rocks and for assessing the moisture content of vegetation.

- Do you think radar imaging could take place at night?
  Yes, because radar carries its own source of illumination (irradiation) and is thus not dependent on scattered sunlight.

- From your knowledge of radio reception and mobile telephones, do you think radar imaging could take place through clouds and rain?
  Yes, certainly at wavelengths corresponding to frequencies of about 1-2GHz. Once radar frequencies exceed about 10GHz, however, rain attenuation can be a problem, although clouds are generally OK. See J.A. Richards, *Radio Wave Propagation*, Springer, Berlin, 2008.

  You may wish to note that $\lambda(m) = {}^{300}/_{f(MHz)}$

Lecture 6. Distortions in recorded images

- Suppose a natural colour photograph of a vegetated region is taken from a satellite. Describe the general appearance of the image under the three scattering conditions illustrated five slides back.
  There will be significant scattering of blue sunlight under all three conditions, so that the blue band would appear hazy; the other two would look moderately free of any atmospheric effect. At the other extreme of particulates (fog, for example) all visible bands would be affected showing a whitish haze over the scene. For the in-between scattering condition, the blue band would look hazy, the green band slightly so, and the red band clear of any major effects.

- Can you think of a simple way to correct line striping caused by mismatches in the input-output characteristics of a detector?
  The histogram matching technique could be used as described in the lectures, in which the histograms of the lines of pixels recorded by each detector are matched.
  A simpler approach is to compute the mean and standard deviation of the lines of pixels recorded by each detector and match the signals of all other detectors to the first one such that the means and standard deviations then become all the same. A formula for doing that will be found on page 30 of J.A. Richards, *Remote Sensing Digital Image Analysis*, *5th Ed.*, Springer, Berlin, 2013.

- What would be recorded in each of the following situations, over the visible to reflected infrared range of wavelengths:

  - No atmosphere and an ideal reflecting surface that reflects all incident energy, irrespective of wavelength
    The surface would appear white

  - A clear atmosphere and an ideal reflecting surface
    The surface would appear white, with some blue haze

  - No atmosphere, but a vegetated surface (use the spectral reflectance curve for vegetation from Lecture Three
    The surface would be coloured according to the positions of the bands on the vegetation reflectance spectrum

  - A clear atmosphere and a vegetated surface
    The surface would be coloured according to the positions of the bands on the vegetation reflectance spectrum but with a slight emphasis to the blue end of the spectrum

Lecture 7. Geometric distortions in recorded images

- How would each of the following sources of geometric distortion appear in a displayed image?

  - earth curvature
    The image would appear to roll off at the edges, giving rise to the S-bend effect

  - finite sensor scan time
    The side of the image swath at the end of the scan would appear shifted backwards (in the displayed product)

  - platform roll
    The displayed imaged will be distorted in a rolling fashion (from side to side along the swath) but in the opposite direction to the physical roll

  - platform gradually rising in height during orbit
    The displayed image will look compressed as the track progresses, because a greater swath width is being imaged with time but displayed as the same width as the earlier parts of the scan

  - Would the following be sources of geometric distortion for a push broom scanner?

    - Finite scan time?
      No, because there is no across the track scanning

    - scanning non-linearities
      No, because there is no scanning mirror

    - Wide field of view systems
      Yes, because a wide FOV is a function of the design of the area to be scanned and is not dependent on the type of imaging device

Lecture 8. Correcting geometric distortion

- One form of geometric distortion is a scale change horizontally that results from over- or under-sampling along a scan line. What do you think the correction matrix might look like?
  The recorded image would have its horizontal scale distorted compared with the correct scale on the map. The vertical scale is unaffected. To correct the error, we can scale either the horizontal (x) axis or the vertical (y) axis to get the correct aspect ratio. In practice the vertical scale is usually adjusted. For example, in the

case of the Landsat multispectral scanner the vertical scale is expanded by 1.411 to account for the over-sampling along a scan line (see J.A. Richards, *Remote Sensing Digital Image Analysis*, 5th ed., Springer, Berlin, 2013, page 65.)

Thus: $x = u$ and $y = av$. In matrix form this distortion is described by $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$. To correct the image, we have to compute the map coordinates from the image coordinates, which means inverting this expression to give $\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1/a \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$.

- Show how the corrections for several different distortions can be combined into a single step.
  Let $\mathbf{M}1, \mathbf{M}2, \mathbf{M}3$ etc. be matrices that correct for different types of geometric distortion. Then the combined distortion can be described by $\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{M}_1\mathbf{M}_2\mathbf{M}_3 \begin{bmatrix} u \\ v \end{bmatrix}$, which needs to be inverted to produce the correction equation.

- Later in this course we are going to look at methods for deciding what ground cover type is represented by each pixel. We will then give the pixel a label. Do you think it would better to do that before removing geometric distortions, or should the geometry be corrected beforehand?
  In principle it shouldn't really matter, especially if nearest neighbor resampling is used (see later lecture). However, there is a school of thought which says that is it better not to do anything to the original pixel values that might introduce noise or some small difference into their values. In that case, it would be better to analyse the original data to create class labels beforehand and then carryout the correction of geometry.

Lecture 9. Correcting geometric distortion using mapping functions and control points

- In this lecture we have set up polynomials that find the image coordinates of a pixel for a given map position. Is it possible to do the reverse – i.e. find map coordinates from the pixel position in an image?
  Yes, by expressing the map coordinates (x and y) as polynomials in the image coordinates (u and v).

- Compare the benefits and limitations with using very high degree mapping polynomials as against first order mapping polynomials.
  High order or high degree polynomials will fit the control points but will not extrapolate well beyond the region of the control points. By comparison lower order polynomials will extrapolate acceptably but may not fit as well in the vicinity of the control points.

- Can you see how the technique used in this lecture could be used to register two images together?
  One image (called the master) can take the place of the map, and the other image (called the slave) can be registered to it. If, however, it is important that the registered pair have good alignment with a map base, it is better to register each of them to the map separately.

Lecture 10. Resampling

- What would guide you in selecting good control points?
  Points that are well defined geographically on both the map and the image and not liable to change with time (as with many water features).

- Is it preferable to use mapping polynomials of low or high order?
  Generally higher order if one has confidence that the control points are accurately located **and** are well spread over the image.

- Nearest neighbor resampling uses recorded pixel brightness values for placement on the map grid, whereas cubic convolution resampling creates new, interpolated brightness values. Is either preferable?
  Cubic convolution is generally preferable if there is a difference in the scales of the map and image and to ensure that the registered image is smooth looking. Nearest neighbor is good if the scales of the image and map are not too different.

- Could the registration techniques treated in the last two lectures be used to register one map to another map?
  Yes, provided the maps are digitized and it is possible to recognize sets of control points in both. Nearest neighbor resampling is then the only viable resampling option.

Lecture 11. An image registration example

- When might you prefer nearest neighbor resampling over cubic convolution resampling?
  When registering a map, such as when the pixels of an image have been given labels indicating what they represent on the ground, or when speed is important in resampling.

- When registering to a map grid is the relative scale of the map and the image important?
  The registered image will be better if the scales are comparable. Registration can still be carried out is the scales are quite different, but care must be taken not to use nearest neighbor resampling.

- When registering two images is the relative scale of the images important?
  As in the previous question, provided a higher order resampling technique (cubic convolution) is used.

- Suppose you have to register five images together to create an image database in a GIS. Would you choose one image as a master and then register the other four to it, or would you register image 2 to image 1 and then image 3 to the newly registered image 2, and so on?
  It is better to choose one as a reference and register all the others to it. Otherwise geometric errors will propagate with each successive registration.

Lecture 12. How can images be interpreted and used?

- Would a human photointerpreter generally examine an image at the individual pixel level?
  It could be done but, given the number of pixels in an image, it would be impracticable.

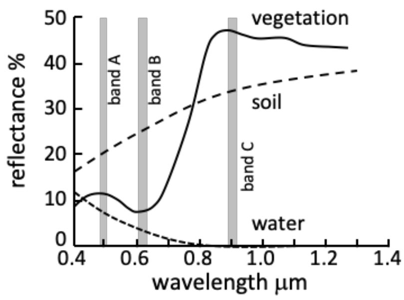
- A 1000x1000 pixel image has 1 million pixels. Even if the analyst chose to look at individual pixels why would that not be effective?
  Having to look at 1,000,000 pixels is not realistic. The analyst could not guarantee to use the same level of concentration on each pixel in such a large numbers.

- Discuss the limitations on human interpretation of an image that contains 100 bands, with each pixel having a radiometric resolution of 8 bits.
  Again, it is impractical to think that an analyst could look at the properties of pixels over 100 different bands. Further, a radiometric resolution of 8 bits means each pixel has 256 brightness values from black to white; a human can generally only discern about 15-20 different levels of grey.

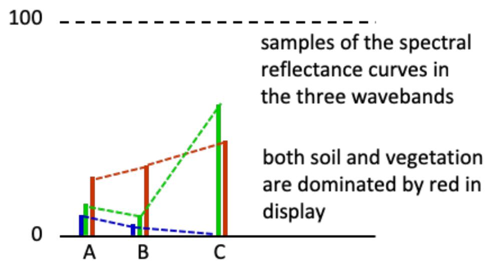
- What sorts of features are best recognized by a human interpreter?
  Spatially-defined features such as shapes, lines and edges.

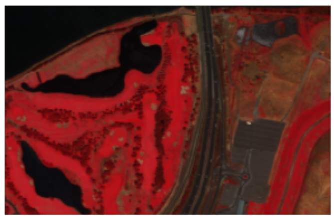
Lecture 13. Enhancing image contrast

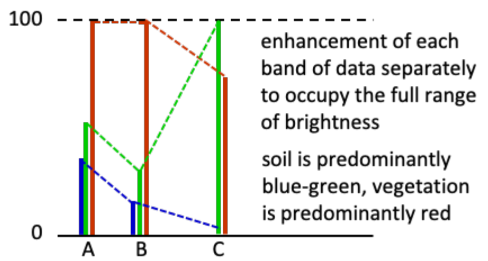
- Explain why all cover types appears reddish in the as-recorded example of this lecture whereas in the enhanced version soils and bare regions appear blue and bluish-green.
  This is best illustrated in a diagram, as below:

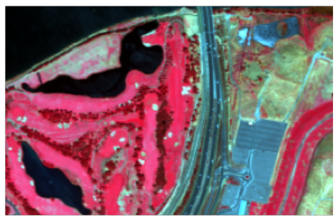


band	display as
C	red
B	green
A	blue









- The brightness value mapping function used in this example is linear. Can you suggest other (non-linear) mapping functions that might be particularly useful?
  There is a whole range of possibilities. Piecewise-linear functions can be used to control brightness changes in certain ranges, an exponential function will highlight differences in the higher brightness levels, whereas a logarithmic function will highlight differences at lower brightness levels. See Section 4.3 in J.A. Richards, *Remote Sensing Digital Image Analysis*, *5th Ed.*, Springer, Berlin, 2013.

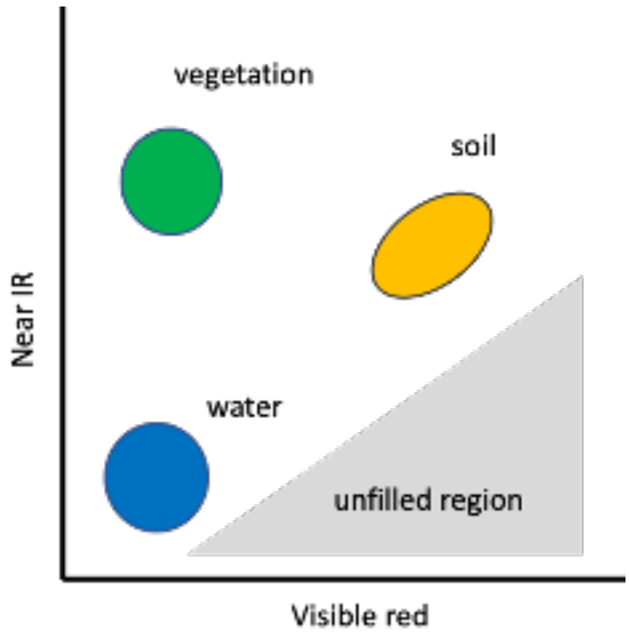
- Suppose a particular sensor has two wavebands. We can construct two individual histograms, in which the independent variables (abscissas) are the brightness values in each waveband. For this image can a histogram be constructed with two independent variables?

Yes, it will be a two-dimensional array of cells, with the axes corresponding to the brightness values in each of the two bands and the cell values being equal to the count of those pixels which have both of the brightness values.

Lecture 14. An introduction to classification (quantitative analysis)

- In two dimensions, how do you think we might find the lines that separate the three classes in the example in this lecture?
  A simple approach is to find the mean vector of each class and then choose lines that are at right angles to, and half way along, the lines between pairs of class means. The collection of those three lines will enable the three classes to be separated. Effectively, that process labels a pixel according to which is the closest class mean.

- Thinking carefully about the spectral reflectance curves of vegetation, soil and water, where would pixels of each of those cover types appear in a two-dimensional spectral space in which the brightness in a near infrared band is plotted vertically, while the brightness in a visible red band is plotted horizontally?



- Would there be parts of that spectral space which would always be largely empty?
  See the diagram above. There are no natural cover types with low IR and high visible red response.

- Where would deep shadows appear in a spectral space?
  Near the origin.

Lecture 15. Classification: some more detail

- How many individual pixel points are there in the spectral space for an image with 100 spectral bands and an 8 bit radiometric resolution?
  8 bit radiometric resolution allows for 256 different levels of brightness (black to white). Thus, for a 100 dimensional space there will be 256^{100} individual cells (pixel points) in the spectral space.

- For each of the following applications would photointerpretation or classification be the most appropriate analytical technique?
  - creating maps of land use
    classification
  - mapping the movement of floods
    either, depending on the spatial accuracy required
  - determining the area of crops
    classification
  - structural mapping in geology
    photointerpretation
  - assessing forest condition
    classification
  - mapping drainage patterns
    photointerpretation
  - creating bathymetric charts
    classification

Lecture 16. Correlation and covariance

- Is a high degree of correlation between bands in a remote sensing image good or not?
  The higher the correlation the less additional information, on the average, is provided by each band. A fully un-correlated set of bands, which is highly unlikely, means all are required to describe the landscape being imaged.

- If two of the bands in an image were 100% correlated, what does that tell us about the value (or otherwise) of having both of those bands in our image data set?
  This means that one band is redundant and can be ignored.

- Look carefully at the correlated data set we used for the examples in this lecture. Can you find a different set of axes in which the data would appear uncorrelated?

  An axis which was along the direction of the dotted green line, with another at right angles to it, would provide a coordinate system in which the set of data points would appear to be uncorrelated.

- What will be the dimensionality of the covariance and correlation matrices for images recorded by:
  - The Landsat ETM+ scanner (ignore the panchromatic band).
    7
  - The original SPOT HRV sensor.
    3
  - The Sentinel-3 OLCI instrument
    21
  - The EO-1 Hyperion sensor.
    220

Lecture 17. The principal components transform

- In this lecture we have looked at rotating the coordinate system to reduce correlation. What would happen to correlation among the bands if we shifted the origin of the coordinate system instead?
  The correlation would be unaffected since shifting the origin does not entail a rotation of the coordinates.

- Is $\mathbf{G}^{\mathrm{T}}\mathbf{G}$ the same as $\mathbf{G}\mathbf{G}^{\mathrm{T}}$, where $\mathbf{G}$ is a matrix?
  No, matrix multiplication is not commutative; the order of the product is important.

- What about $\mathbf{g}^{\mathrm{T}}\mathbf{g}$ and $\mathbf{g}\mathbf{g}^{\mathrm{T}}$ where $\mathbf{g}$ is a column vector?
  Likewise, vector multiplication is non-commutative; the order is important, as the following shows:

$$\mathbf{g}^{\mathrm{T}}\mathbf{g} = [a \quad b \quad c]\begin{bmatrix} a \\ b \\ c \end{bmatrix} = a^2 + b^2 + c^2 = \text{scalar}$$

$$\mathbf{g}\mathbf{g}^{\mathrm{T}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}[a \quad b \quad c] = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix} = \text{matrix}$$

To answer these last two questions you might wish to use the following simple examples:

$$\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \qquad \mathbf{g} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Lecture 18. The principal components transform: worked example

- In a two-dimensional case, if one of the eigenvalues was zero, what does that say about the spread of the pixel vectors in the original spectral space?
  They must lie along a straight line—in other words the bands are fully correlated.

- If the two eigenvalues were about the same what does that tell you about the original data distribution?
  It is roughly circular or squashed circular in two dimensions—in other words the two dimensions are totally in-correlated.

- If one of the transformed bands (i.e. principal components) had a very small variance, does that mean that that band does not contain very much useful information?
  Not necessarily. The covariance matrix for an image, from which the principal components transformation derives, is an average measure—recall the expectation operator—so that small details may have little influence in the computation. As a result, we often find significant small geometric features and classes in later, low variance, principal components.

Lecture 19. The principal components transform: a real example

- For the two examples given in this lecture, using the values of the eigenvectors in each case, describe how much of each of the original bands contributes to the first and fourth principal components.

First data set, first eigenvector	0.22	0.26	0.72	0.61
First data set, fourth eigenvector	0.68	-0.70	-0.10	0.17

Second data set, first eigenvector	0.47	0.49	0.52	0.52
Second data set, fourth eigenvector	0.25	0.04	-0.80	0.54

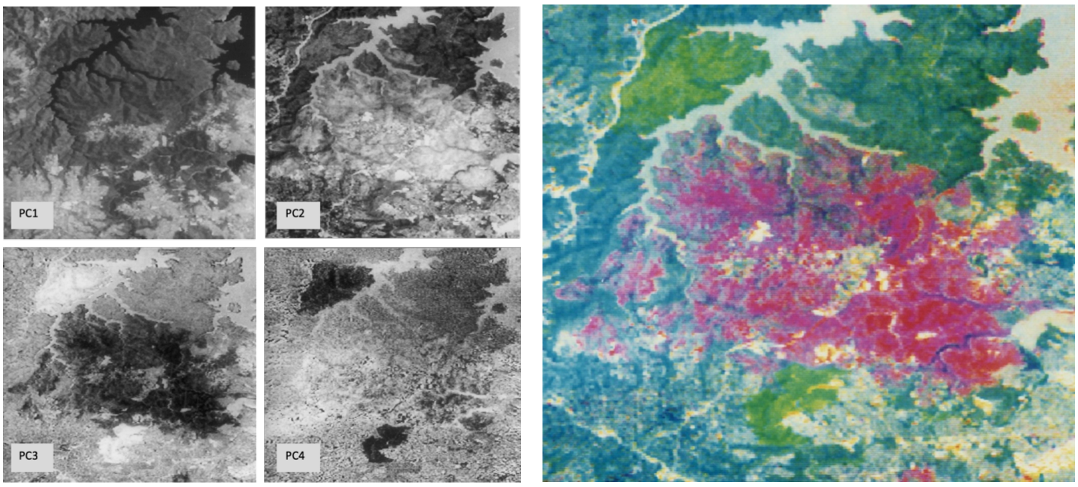
In the first case the first eigenvalue is dominated by the last two elements, meaning that bands 29 and 80 are contributing most to the first PC. Nevertheless, the other bands also make significant contributions; all four bands contribute additively. The four eigenvector is dominated by the original bands 7 and 15. Since their signs are different the fourth PC will reflect differences between those original bands.

In the second case the first PC is made up of almost equal proportions of the original four bands. The fourth largely shows up the difference between band 40 and a weighted sum of bands 30 and 45, because of the sign differences.

- By inspecting the eigenvalues of the covariance matrix of an image could you decide beforehand whether it is worthwhile proceeding to carry out the principal components transformation?
  In principal yes. The greater the fall-off in the sizes of the eigenvalues the greater effect a principal components transformation will have.

Lecture 20. Applications of the principal components transform

- Explain the colours in the PC colour composite image of the bush fire example.



  The colour composite is made up of PC2, PC3 and PC4, displayed respectively as red, green and blue. The central 1979 bushfire scar is a dominant light feature in PC2, that's why is shows as mainly red in the colour image. The two 1980 bushfire scars are low in brightness in PC2 and PC4, but light in appearance in PC3; thus, they are predominantly green in the colour version. Water features are light in all three PCs so it shows as a grey shade in the colour image. The unburnt vegetation is dark in PC2, light in PC3 and mostly dark in PC4, except for the peninsula to the top right. Consequently, that cover type is dominated by blue-ish green, except for the peninsula where green dominates.

- Do you always have to compute the covariance matrix of the original pixel vectors over the full image?
  No, it can be computed over any convenient part of the image, in order to emphasise particular features of interest in a principal components transform.

- When might you want to use a subset of the image to generate a covariance matrix for principal components transformation?
  As above, when trying to emphasise differences in some cover types, and where the other cover types are less important.

- Would your answer to the last question help if the temporal changes between two images were comparable in size to the portion of the scene which had not changed between dates?
  It is hard to predict where the principal axes will fall in a multi-temporal spectral space when the change region is a large proportion of the scene. But if the covariance matrix is chosen by focussing just on the parts of the scene that have changed, the first PC, or first few, are likely to be dominated by the change event. See the following simple variation of the fire example treated in this lecture. Here the two change events (dark green and red) are much more dominant. On the left-hand side we assume the covariance matrix has been compiled over the whole scene, whereas in the second it has been compiled just using the pixels which have shown change.